

Juan David Muñoz Bolaños

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IMS Nanofabrication GmbH

Recruiting Team

Dresdner Str. 47, 1200 Vienna, Austria

Application for the position: Development Engineer

Dear Recruiting Team,

IMS Nanofabrication drives the limits of lithography at the nanometer scale. As an optical systems engineer finishing my PhD at the Medical University of Innsbruck, I have spent the last four years building complex optomechatronic systems (advanced microscopes) and developing data analysis pipelines to optimize their performance. My background in physics engineering, combined with my exposure to industrial semiconductor workflows at the ASML and ZEISS summer schools, makes me eager to contribute to the development of your Multi Beam Mask Writer systems.

In my current research, I designed and built an automated wavefront sensing and adaptive optics system from scratch. This involved integrating high precision optical components, spatial light modulators, and scientific cameras, while writing the entire control and data acquisition stack in Python and C++. To ensure the system achieved the best potential performance, I developed iterative phase retrieval and optimization algorithms to process wavefront data in real time. This hands-on experience in combining hardware control with advanced signal processing directly aligns with your need for validating new features and analyzing complex log and metrology data.

Beyond system development, I focus on structured testing and quality frameworks. I have structured my instrument development following V-Model systems engineering principles, ensuring clear requirements tracking from initial concept to final validation. By applying Design of Experiments and systematic optical tolerance analysis in python, I have successfully bridged the gap between early stage R&D and stable, reproducible system performance.

I will complete my PhD in June 2026. I am Colombian originally and hold an unrestricted working resident permit in Austria (rot-weiss-rot-plus). I look forward to discussing how my background in optomechanical and electronic systems, data analysis, and structured testing can support your team in Vienna.

Best regards,

Juan David Munoz Bolanos

Juan David Munoz Bolanos

Junior Optical System Engineer

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Summary

Junior system engineer with hands-on lab work building, aligning, and calibrating optical laser systems using spatial light modulators and holography. I'm comfortable with PC and control technology, used to digging into technical problems systematically.

Research/Professional Experience

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Optical Systems Engineer / PhD Researcher

Optomechatronic System Design: Designed and built a multiphoton microscope and an automated wavefront sensing system from scratch, integrating lasers, spatial light modulators, and deformable mirrors.

Data Analysis & Algorithms: Developed Python and C++ pipelines for real time signal processing, iterative phase retrieval, and wavefield simulations to optimize system performance.

Systems Engineering & Quality: Applied V Model methodologies, Design of Experiments, and FMEA to track requirements and systematically resolve technical defects during prototype iterations.

Lab Management & Safety: Managed a high power laser laboratory, ensuring EH&S compliance, maintaining equipment logs, and coordinating component sourcing with suppliers like Thorlabs and Edmund Optics.

Aug 2018 to Oct 2019 **INTECOL S.A.S.** *Medellín, Colombia*
Junior Machine Vision Engineer (Field Service)

Machine Vision Pipelines: Engineered and deployed Halcon and Cognex VisionPro image processing pipelines for high speed industrial production lines, improving quality control.

Illumination Design: Designed custom bright field and dark field illumination systems to reliably detect defects, liquid levels, and sealing states on fast moving products.

Industrial Automation: Integrated vision systems with robotic arms and PLC controllers, executing on site testing, validation, and customer sign off.

Technical Competencies

Software & Tools

Python C++ LabVIEW Halcon OpenCV ZEMAX MATLAB Git

Optical Systems

Adaptive Optics Wavefront Sensing Microscopy Spectroscopy Laser Systems
DMDs

Systems & Quality

V-Model 8D Problem Solving Design of Experiments ISO 9001

Education

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Doctor of Philosophy in Optical Metrology

Oct 2019 to Sep
2021

Friedrich Schiller Univ. Jena
Master of Science in Photonics

Jena, Germany

Aug 2012 to Aug
2018

University of Cauca
Bachelor of Science in Physics Engineering

Popayán, Colombia

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

Biking/Hiking Bachata Dancing Harmonica Learning German

Languages

English (Fluent) German (Intermediate B2) Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
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PD Dr. Christoph Krafft
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